

BD Platform — Technical Overview

The as-built architecture: how opportunity intelligence flows from ~100+ public sources through canonical identity resolution into pursuits, CRM, and daily reporting — and the enforcement machinery that keeps it from drifting.

Stack .NET 8 · SQL Server · Quartz · WPF Worker 1.0.9688.1083 live As of 2026-07-11

TOPOLOGY

Three machines, one database of record

DEV · KOR-1001

Build & deploy origin

- Repo `Operations` (git, branch `develop`)
- `publish-opportunities.ps1` — test-gated publish
- Supervised CLIs (dedup, imports, drains)
- Push deploys via `Invoke-Command` + robocopy



SERVER · KOR-APP01

Runtime core

- **Kor.Opportunities.Worker** — ~30 Quartz jobs + pollers (service, runs as `KOR\app-admin`)
- **SQLEXPRESS · KorOpportunitiesDb** — schema `opportunities`, the system of record
- MCP server (`/ask` text-to-SQL, :5500)
- FileSync service (separate product)
- Deltek Vantagepoint via ODBC + linked server `[DELTEK_VP]`



WORKSTATIONS

Consumption

- WPF app (NewerForma, V15 zip) — BD workspace, CRM, dossiers, reports
- `kor://` deep links from PDFs/email into the running app
- 6 a.m. morning email (manager + per-owner digests)

LAN-only. All automation lives on KOR-APP01; nothing schedules on workstations.

DATA FLOW

Intake → resolve → repair → consume

1 · INTAKE

Two scheduling planes, one dispatch spine

~100+ registered sources (`OpportunitySources`): BC Bid, CanadaBuys, SAM.gov, 32 Bonfire tenants, bids&tenders, CivicInfo RSS, municipal permit feeds, news feeds, a Graph-pollled alerts mailbox, and four Major-Projects-Inventory providers (AB/BC/CA×2). Named Quartz jobs own the big feeds; a generic cron scheduler ticks every 5 min for the rest. Plane ownership is **data** (`OpportunitySources.QuartzManaged`), never a code list; failed sources back off a full crawl window; a provider-name miss throws instead of running the wrong parser.

2 · GATE & DEDUP

Relevance gate, then content hashing

`StructuralRelevanceGate` word-bounded keep/reject vocabularies (suffix-aware so *constructing/towers* match but *infrastructure/additional* can't false-pass); rejects are recorded for tuning. Candidate identity = SHA-256 over Title|Buyer|Location|Url|Deadline against `OpportunityObservations` ; upsert keys on source-prefixed external reference.

3 · CANONICAL IDENTITY

One org, one row — enforced physically

- `CanonicalOrgResolver`: 6-tier deterministic match (alias → merged-survivor redirect → strict normalized name → domain → fuzzy → create), junk denylist + concatenated-entity guard, `TeamNameCleaner` at every front door.
- **Lifecycle contract**: firehose feeds (awards/bids/permits) reuse archived identity *cold*; deliberate callers resurrect dormancy-archived rows; curation retirements are final; dedup merge-losers can never come back.

- **Physical guarantee:** unique filtered index — one live org per normalized name. Twins are impossible, not just guarded.
- Merging is supervised-CLI only (`BdCanonicalDedup`), with a merge ledger, survivor redirect, and person-preserving repoints.

Person identity mirrors this: `usp_ResolveOrCreateIntelPerson` anchors on email → LinkedIn → name+org, resurrects retired people on re-discovery, refuses bare-name creation, and gates guessed/shared emails out of identity matching.

4 • WRITE DOCTRINE

Fill-only, name-and-FK together

Every scheduled writer follows one rule set: canonical FKs are fill-only (a resolver miss can never null a good link); a name and its FK move as a pair (a row can never display one firm while linking to another); curated/enriched values survive raw-feed refreshes. Applies uniformly across the four MPI providers, the Deltek pursuit sync, the bid store, and the research importer.

5 • REPAIR WHEEL

Write-path-agnostic self-healing

`CanonicalColumnRegistry` declares all 18 (table, name-column, FK-column) triples that reference `CanonicalOrg` — verified against live schema at startup, **fail-closed** (drift = Worker refuses to start). Three consumers share it: `CanonicalLinkBackfillJob` (weekly keyset-paginated pass that links any name-set/FK-null row via read-only `FindExistingAsync` — never mints, never overwrites), the health audit (measures the same columns), and the dedup tool (repoints the same FKs on merge). Adding a canonical column is one registry entry; forgetting it is a loud failure.

6 • BUSINESS LIFECYCLE

Opportunities → pursuits → CRM → outcomes

- **Opportunities:** tender rows with buyer canonical links, grab/claim concurrency for owner assignment, duplicate guard on manual creation.
- **MPI / forward pipeline:** 2,600+ major projects with team columns (architect/SE/GC/proponent), seat-status vocabulary (`likely-open` etc.), on-demand honing synthesis with per-verdict TTLs, sector dossiers built live.
- **KorPursuits + CRM:** Deltek PR sync (one pursuit = one row; won transitions detected via `~WDEF~` sweep since Deltek has no Won stage), engagements with stage history, and Won/Lost **write-through** to the opportunity ledger in the same transaction — one answer to "did we win?" everywhere.
- **Deltek fusion:** clients via `Clendor`, spend/fees via ODBC; Deltek owns financials, CRM owns outcomes.

- **Retirement:** nightly archive-not-delete lifecycle (`RetiredAtUtc`) keeps a warm working set without destroying history.

7 · OBSERVABILITY

Measured, delivered, red on failure

DB heartbeat every 60 s (authoritative liveness, carries the running version). Jobs rethrow on failure so run history shows red. The 6 a.m. morning email carries: new opportunities, verdict movement, owner digests, coverage, **DEAD-GREEN source health** (enabled sources with no successful run inside 3× their crawl window), drain-queue status, and summary flags. The weekly health audit measures registry orphan rates with name-aware denominators.

ENFORCEMENT

The doctrine layer — why fixes stop drifting

Ten binding invariants (`docs/BD-Doctrine.md`), each traced to the mechanism that enforces it. The architecture tests scan the actual source tree; escapes require a written reason in a reviewable allowlist.

GUARD	INVARIANT	FAILS HOW
D1 test	No unguarded canonical-FK SQL writes anywhere in the tree	Red build
D2 test	Every <code>*CanonicalOrgId</code> column registered for repair or consciously excluded	Red build
D3 test	Every Quartz job visible in the Admin schedule registry	Red build
D4 test	The junk-placeholder corpus can never survive the cleaner again	Red build
D5 runtime	Registry matches live schema	Worker refuses to start
D6 schema	One live org per normalized name	Physically impossible (unique index)
D7–D8 proc/tier	Person + org lifecycle contracts (resurrect / keep-cold / curation-final)	Pinning tests
D9 data	Scheduling ownership is a column, not a code list	Migration-stamped
D10 delivery	Failures red, rot emailed	Morning report

The gate: there is no CI server, so the deploy pipeline is the enforcement point — `publish-opportunities.ps1` runs the full test suite (59 tests: doctrine + behavioral) and **refuses to produce a deployable artifact on red**. Unverified code structurally cannot reach KOR-APP01.

OPERATIONS

Deploy & verify

STEP	MECHANISM
Publish	Test gate → <code>dotnet publish</code> → timestamped artifact with <code>FileVersion 1.0.<days-since-2000>.<minutes></code> + git SHA in ProductVersion; last 3 retained
Deploy (Worker)	From KOR-1001: stop service (poll to Stopped + 8 s handle buffer) → robocopy /MIR → start → verify FileVersion decodes to <i>today</i>
Verify	Process version + DB heartbeat (authoritative), startup log line <code>CanonicalColumnRegistry: verified 18 canonical columns</code>
App	Versioned V<N>.zip to the firm share (playwright-culled, add-in grafted, SHA-verified, prior zip = rollback); workstations install manually. Packaging timing is an operator decision
DB migrations	Numbered <code>Schema/*.sql</code> , connection-scoped (no USE), applied with <code>QUOTED_IDENTIFIER ON</code> ; applied-state noted in the file header

HONEST LEDGER

Known gaps, named

STATUS	ITEM	POSTURE
LIVE	Dead-man's switch — independent scheduled task on KOR-APP01 checks heartbeat age every 10 min; auto-restarts a stopped service; alerts by email (1 h cooldown) + recovery notice	<code>Watch-OpportunitiesHeartbeat.ps1</code> · verified end-to-end 2026-07-11
OPEN	Plaintext secrets (machine env vars / appsettings); rotation pending	Acceptable LAN-only; below product bar

ACCEPTED	No staging environment; single-instance SQLEXPRESS; DR = server backups	Standard internal-tool posture
ACCEPTED	Fuzzy tier has no confidence score / kind-compat check	Bounded: read-only wheel pass can't mint; batch volume monitored
LIVE	Everything else in this document	Worker 1.0.9688.1083 · migrations 277–281 applied · 59/59 tests green

KOR BD Platform · engineering reference · generated 2026-07-11 · sources: repo `Operations@develop`, live KOR-APP01 verification, docs/BD-Doctrine.md

Companion documents: BD-Doctrine.md (binding invariants) · audit remediation plan artifact · BD Handbook / QuickStart (user-facing)